

A Counterexample to Conjecture 00000007990: Perfect Lee Codes of Radius 1 Exist in Every Dimension

Feiyu*

Claude (Opus 5.5)[†]

2026-10-03

Abstract

Conjecture 00000007990 claims that perfect Lee codes $PL(n, e)$ exist only for $(n, e) = (2, 2)$, for $e = 2$ with $n \equiv \pm 1 \pmod{6}$, and for sparse exceptions of higher radius. In particular no perfect Lee code of radius 1 would exist. But for every $n \geq 1$ the Golomb-Welch code $\{x \in \mathbb{Z}^n : \sum_i i x_i \equiv 0 \pmod{2n+1}\}$ is a perfect Lee code of radius 1. The refutation, for all n at once, is checked in Lean 4 against Mathlib.

1 The conjecture as stated

The original text of conjecture 00000007990 reads:

Definition: The perfect covering of the Lee metric for codewords: perfect Lee codes (a continuation of the Golab-Weldon problem). Conjecture: Perfect Lee codes $PL(n, e)$ of radius e exist only at $(n, e) = (2, 2)$ and n congruent to plus or minus 1 mod 6 ($e = 2$, known to Golab-Weldon) plus sparse higher-radius exceptions; the step distribution of (n, e) for radius at least 3 is an arithmetic-progression intersection in the prime-mod-6 sense with density $(1/2) \prod_{p>3} (1 - 2/p^2)$ (an explicit Euler product); and nonexistence is uniquely determined by a Diophantine approximation obstruction of lattice density (the Markov constant).

The Chinese version in `conjectures/00000007990.md` states the same three claims; it confirms that “higher radius” means radius at least 3.

2 Reading of the statement

Definition 1. The Lee distance on $\mathbb{Z}^n$ is $d(x, y) = \sum_{i=1}^n |x_i - y_i|$. A set $C \subseteq \mathbb{Z}^n$ is a *perfect Lee code of radius e* , written $PL(n, e)$, if every point of $\mathbb{Z}^n$ is within Lee distance e of exactly one codeword; equivalently, the Lee balls of radius e around the codewords tile $\mathbb{Z}^n$.

This is the setting of Golomb and Welch [1], whose name the statement garbles as “Golab-Weldon”. The conjecture is a conjunction of three claims. We refute the first one:

(C1) if $PL(n, e)$ exists with $n, e \geq 1$, then $(n, e) = (2, 2)$, or $e = 2$ and $n \equiv \pm 1 \pmod{6}$, or $e \geq 3$.

Allowing every radius $e \geq 3$ in (C1), instead of only “sparse” exceptions, only weakens the claim. Under (C1) no perfect Lee code of radius 1 exists.

*Independent researcher.

[†]Anthropic.

3 Result

Theorem 2. *For every $n \geq 1$, the Golomb–Welch code*

$$C_n = \left\{ x \in \mathbb{Z}^n : \sum_{i=1}^n i x_i \equiv 0 \pmod{2n+1} \right\}$$

is a perfect Lee code of radius 1. Hence (C1) fails for every pair $(n, 1)$, $n \geq 1$, and Conjecture 00000007990 is false.

4 Proof

Write $w(x) = \sum_i i x_i$ and $m = 2n + 1$, and let e_i be the unit vectors. A point y satisfies $d(x, y) \leq 1$ iff $y = x$ or $y = x \pm e_i$ for some i . Indeed, if $d(x, y) \leq 1$ and $y \neq x$, some coordinate differs, contributing at least 1, so it contributes exactly 1 and all other coordinates agree. These $2n + 1$ points have weights

$$w(x), \quad w(x) + i, \quad w(x) - i \quad (1 \leq i \leq n),$$

that is $w(x) + \varepsilon$ with ε running over the $2n + 1$ distinct integers $-n, \dots, n$. These are all the residues modulo $m = 2n + 1$, each exactly once. So exactly one of the points has weight divisible by m , i.e. lies in C_n .

In the formal proof, existence takes the residue $s = w(x) \bmod m \in [0, 2n]$ and steps by 0, by $-e_s$ (if $1 \leq s \leq n$) or by $+e_{m-s}$ (if $n < s \leq 2n$). Uniqueness uses that two admissible steps have weights congruent modulo m and of absolute value at most n , hence equal weights, hence they are equal.

5 Formalization

The Lean 4 project in `lean/` (file `lean/Submission/Basic.lean`) formalizes the definitions above and proves Theorem 2 against Mathlib.

- Points are integer vectors `Fin n → ℤ`; `leeDist` is the Lee distance, and `IsPerfectLeeCode n e C` states that every point has a unique codeword within distance e (Lean’s `ExistsUnique`). `ExistsPL n e` asserts that some such C exists. `Claim1` is (C1).
- `weight` is w (with coordinates indexed from 0, so the weight of coordinate i is $i + 1$), and `GW n` is C_n .
- `step` and `stepWeight` describe the moves $0, \pm e_i$ and their weights. `eq_add_step` characterizes the points within distance 1; `stepWeight_abs` and `stepWeight_injective` are the facts used for uniqueness.
- `isPerfectLeeCode_GW` is Theorem 2 for every n . `not_claim1` refutes (C1), and `claim1_fails_everywhere` records that every pair $(n, 1)$ violates it.
- `conjecture_00000007990_false` states $\neg(\text{Claim1} \wedge P \wedge Q)$ for arbitrary propositions P, Q standing for the other two claims.

The toolchain is `leanprover/lean4:v4.33.1`, Mathlib is pinned to `v4.33.1`, and `lake build` succeeds. The project contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms: `#print axioms` reports exactly `propext`, `Classical.choice` and `Quot.sound` for `conjecture_00000007990_false` and `isPerfectLeeCode_GW`.

6 Remarks

Remark 3. Golomb and Welch [1] showed that perfect Lee codes exist in dimension 1 and 2 for every radius, and for radius 1 in every dimension; their conjecture is that there are no others when $n \geq 3$ and $e \geq 2$. So perfect Lee codes are far more common than (C1) allows; we formalize only the radius-1 family.

References

- [1] S. W. Golomb and L. R. Welch, *Perfect codes in the Lee metric and the packing of polyominoes*, SIAM J. Appl. Math. 18 (1970), 302–317.